

NEXORA_db - SQL TABLES

EXISTING TABLES - SQL reference only

users

```
CREATE TABLE users (  
    id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    email VARCHAR(255) NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    role VARCHAR(20) NOT NULL  
);  
CREATE UNIQUE INDEX ix_users_email ON users (email);
```

products

```
CREATE TABLE products (  
    id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    name VARCHAR(200) NOT NULL,  
    description VARCHAR(2000) NOT NULL,  
    category VARCHAR(100) NOT NULL,  
    price DOUBLE PRECISION NOT NULL CHECK (price >= 0),  
    stock INTEGER NOT NULL CHECK (stock >= 0)  
);  
CREATE INDEX ix_products_name ON products (name);  
CREATE INDEX ix_products_category ON products (category);
```

orders

```
CREATE TABLE orders (  
    id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    user_id INTEGER NOT NULL REFERENCES users(id),  
    total DOUBLE PRECISION NOT NULL,  
    status VARCHAR(20) NOT NULL,  
    created_at TIMESTAMP WITHOUT TIME ZONE NOT NULL  
);  
CREATE INDEX ix_orders_user_id ON orders (user_id);
```

order_items

```
CREATE TABLE order_items (  
    id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    order_id INTEGER NOT NULL REFERENCES orders(id),  
    product_id INTEGER NOT NULL REFERENCES products(id),  
    product_name VARCHAR(200) NOT NULL,  
    quantity INTEGER NOT NULL,
```

```
price DOUBLE PRECISION NOT NULL  
);
```

alembic_version (migration tracking)

```
SELECT version_num FROM alembic_version;
```

NEW TABLES - CREATE TABLE SQL

roles

```
CREATE TABLE roles (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  name VARCHAR(50) NOT NULL UNIQUE,  
  description TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

user_roles

```
CREATE TABLE user_roles (  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  role_id INTEGER NOT NULL REFERENCES roles(id) ON DELETE CASCADE,  
  assigned_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  PRIMARY KEY (user_id, role_id)  
);
```

addresses

```
CREATE TABLE addresses (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  label VARCHAR(50), recipient_name VARCHAR(150) NOT NULL,  
  phone VARCHAR(40) NOT NULL, address_line1 VARCHAR(200) NOT NULL,  
  address_line2 VARCHAR(200), city VARCHAR(100) NOT NULL,  
  region VARCHAR(100), postal_code VARCHAR(30) NOT NULL,  
  country VARCHAR(2) NOT NULL, is_default BOOLEAN NOT NULL DEFAULT false,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

categories

```
CREATE TABLE categories (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  name VARCHAR(100) NOT NULL, slug VARCHAR(120) NOT NULL UNIQUE,  
  parent_id INTEGER REFERENCES categories(id) ON DELETE RESTRICT,  
  description TEXT, is_active BOOLEAN NOT NULL DEFAULT true  
);
```

brands

```
CREATE TABLE brands (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  name VARCHAR(120) NOT NULL UNIQUE, slug VARCHAR(140) NOT NULL UNIQUE,  
  description TEXT, is_active BOOLEAN NOT NULL DEFAULT true  
);
```

product_catalog_details

```
CREATE TABLE product_catalog_details (  
  product_id INTEGER PRIMARY KEY REFERENCES products(id) ON DELETE CASCADE,  
  category_id INTEGER NOT NULL REFERENCES categories(id) ON DELETE RESTRICT,  
  brand_id INTEGER REFERENCES brands(id) ON DELETE SET NULL,  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

product_images

```
CREATE TABLE product_images (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE CASCADE,  
  image_url TEXT NOT NULL, alt_text VARCHAR(250),  
  sort_order INTEGER NOT NULL DEFAULT 0, is_primary BOOLEAN NOT NULL DEFAULT false  
);
```

carts

```
CREATE TABLE carts (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  status VARCHAR(20) NOT NULL DEFAULT 'active'  
  CHECK (status IN ('active', 'converted', 'abandoned')),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

cart_items

```
CREATE TABLE cart_items (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  cart_id INTEGER NOT NULL REFERENCES carts(id) ON DELETE CASCADE,  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE RESTRICT,  
  quantity INTEGER NOT NULL CHECK (quantity > 0),  
  added_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (cart_id, product_id)
```

```
);
```

wishlists

```
CREATE TABLE wishlists (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  name VARCHAR(100) NOT NULL DEFAULT 'Favorites',  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (user_id, name)  
);
```

wishlist_items

```
CREATE TABLE wishlist_items (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  wishlist_id INTEGER NOT NULL REFERENCES wishlists(id) ON DELETE CASCADE,  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE CASCADE,  
  added_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (wishlist_id, product_id)  
);
```

payments

```
CREATE TABLE payments (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  order_id INTEGER NOT NULL REFERENCES orders(id) ON DELETE RESTRICT,  
  amount NUMERIC(12,2) NOT NULL CHECK (amount >= 0),  
  currency CHAR(3) NOT NULL DEFAULT 'USD',  
  status VARCHAR(20) NOT NULL, provider VARCHAR(60) NOT NULL,  
  provider_reference VARCHAR(200),  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (provider, provider_reference)  
);
```

shipments

```
CREATE TABLE shipments (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  order_id INTEGER NOT NULL REFERENCES orders(id) ON DELETE RESTRICT,  
  status VARCHAR(30) NOT NULL DEFAULT 'pending', carrier VARCHAR(100),  
  tracking_number VARCHAR(150), shipped_at TIMESTAMPTZ, delivered_at TIMESTAMPTZ,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

reviews

```
CREATE TABLE reviews (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE CASCADE,  
  rating SMALLINT NOT NULL CHECK (rating BETWEEN 1 AND 5),  
  title VARCHAR(200), body TEXT,  
  status VARCHAR(20) NOT NULL DEFAULT 'published',  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (user_id, product_id)  
);
```

coupons

```
CREATE TABLE coupons (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  code VARCHAR(60) NOT NULL UNIQUE,  
  discount_type VARCHAR(10) NOT NULL CHECK (discount_type IN ('percent', 'fixed')),  
  discount_value NUMERIC(12,2) NOT NULL CHECK (discount_value > 0),  
  starts_at TIMESTAMPTZ, expires_at TIMESTAMPTZ,  
  minimum_order_total NUMERIC(12,2) NOT NULL DEFAULT 0 CHECK (minimum_order_total >= 0),  
  maximum_uses INTEGER CHECK (maximum_uses IS NULL OR maximum_uses > 0),  
  is_active BOOLEAN NOT NULL DEFAULT true,  
  CHECK (expires_at IS NULL OR starts_at IS NULL OR expires_at > starts_at)  
);
```

coupon_usage

```
CREATE TABLE coupon_usage (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  coupon_id INTEGER NOT NULL REFERENCES coupons(id) ON DELETE RESTRICT,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE RESTRICT,  
  order_id INTEGER NOT NULL REFERENCES orders(id) ON DELETE RESTRICT,  
  discount_amount NUMERIC(12,2) NOT NULL CHECK (discount_amount >= 0),  
  used_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  UNIQUE (coupon_id, order_id)  
);
```

inventory

```
CREATE TABLE inventory (  
  product_id INTEGER PRIMARY KEY REFERENCES products(id) ON DELETE CASCADE,  
  quantity_on_hand INTEGER NOT NULL DEFAULT 0 CHECK (quantity_on_hand >= 0),  
  quantity_reserved INTEGER NOT NULL DEFAULT 0 CHECK (quantity_reserved >= 0),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  CHECK (quantity_reserved <= quantity_on_hand)  
);
```

inventory_transactions

```
CREATE TABLE inventory_transactions (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE RESTRICT,  
  quantity_change INTEGER NOT NULL CHECK (quantity_change <> 0),  
  reason VARCHAR(100) NOT NULL, reference_type VARCHAR(50), reference_id INTEGER,  
  created_by INTEGER REFERENCES users(id) ON DELETE SET NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

notifications

```
CREATE TABLE notifications (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,  
  type VARCHAR(50) NOT NULL, title VARCHAR(200) NOT NULL, message TEXT NOT NULL,  
  payload JSONB, read_at TIMESTAMPTZ,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

product_attributes

```
CREATE TABLE product_attributes (  
  id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
  category_id INTEGER REFERENCES categories(id) ON DELETE CASCADE,  
  name VARCHAR(100) NOT NULL,  
  data_type VARCHAR(20) NOT NULL CHECK (data_type IN ('text', 'number', 'boolean', 'json')),  
  unit VARCHAR(30), is_required BOOLEAN NOT NULL DEFAULT false,  
  UNIQUE (category_id, name)  
);
```

product_attribute_values

```
CREATE TABLE product_attribute_values (  
  product_id INTEGER NOT NULL REFERENCES products(id) ON DELETE CASCADE,  
  attribute_id INTEGER NOT NULL REFERENCES product_attributes(id) ON DELETE CASCADE,  
  value JSONB NOT NULL,  
  PRIMARY KEY (product_id, attribute_id)  
);
```

audit_logs

```
CREATE TABLE audit_logs (  
  -- Table structure for audit_logs
```

```
id INTEGER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
user_id INTEGER REFERENCES users(id) ON DELETE SET NULL,  
action VARCHAR(100) NOT NULL, entity_type VARCHAR(100) NOT NULL,  
entity_id INTEGER, details JSONB NOT NULL DEFAULT '{}'::jsonb,  
created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```